

Reducing Patient Movement in MRI: A Pilot Study of a Training System

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Abstract

The aim of this study is to prepare children and adults for real magnetic resonance imaging (MRI) procedures using specialized training application. The software, designed to track movements and reproduce the characteristic sounds of the procedure, uses the Farneback optical flow algorithm to precisely detect even the slightest movements of participants. This allows for an assessment of the movements during an MRI scan training session in order to improve the patient's performance in real-life scanning. Our study demonstrates that simulator training improved participants' understanding of the procedure. Following the training, 40% of participants showed a decrease in the number of detected movements compared to the baseline test, and 33% of participants did not exhibit any movements in the subsequent test. These results demonstrate that the suggested training can effectively prepare patients for the MRI procedure.

CCS Concepts

• **Human-centered computing** → **Interaction design process and methods**; **Usability testing**; • **Software and its engineering**;

Keywords

MRI Simulator, Children, Preparation, Motion Tracking

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1 Introduction

Magnetic resonance imaging (MRI) is currently a leading method for examining the human body, allowing the detection of diseases at early stages before symptoms appear. This method is considered completely safe, even during pregnancy. MRI scan results are usually available within 24 hours [6], and there are no dietary or drinking restrictions required before the procedure.

MRI diagnostics are particularly important for examining children, as it is the safest and most effective method for detecting congenital pathologies and central nervous system diseases [3, 9]. Despite its advantages, parents often refuse MRI for their children due to the high levels of stress caused by the noise and confined space inside the scanner, which can induce anxiety in children [8]. To obtain high-quality images, complete stillness is required throughout the entire scanning time, which can last from 15 to 60 minutes [5]. This often necessitates the use of sedatives or anesthesia, which carries risks such as respiratory failure and cardiovascular disorders, as well as significant costs [4].

Maintaining stillness during an MRI scan is crucial for obtaining high-quality images. Movement can cause artifacts that degrade image quality and hinder accurate diagnosis. Movement artifacts can blur the image and lead to incorrect interpretation of the results. Optimizing the scanning protocol to reduce the impact of patient movement and enhancing patient compliance through training can significantly improve image quality and diagnostic value. Proper patient preparation, including familiarization with MRI conditions and emphasizing the importance of stillness, can help minimize these artifacts and increase the overall effectiveness of the MRI procedure [7].

Effective feedback mechanisms are essential for the successful conduct of MRI scans, helping patients understand and correct their movements during the procedure. These include visual, auditory, and tactile signal systems that inform the patient of the need to remain still. Visual systems can show on a screen or through real-time projections how still the patient is. Auditory signals can notify the patient of excessive movements, and tactile devices can provide gentle vibrations upon movement. These methods help patients better understand their movements and maintain stillness, significantly improving image quality and reducing the number of motion-related artifacts.

Psychological and emotional preparation for the test is also important. Methods to reduce anxiety before MRI are actively discussed to minimize the need for pharmacological interventions [1]. One way to reduce anxiety in children is the use of virtual reality (VR) [11]. The introduction of VR glasses allowed children to become more familiar with the process, positively impacting their emotional state. VR videos have shown a significant reduction in anxiety levels in children aged 4 to 15 years [2].

The use of models and training platforms for MRI confirms the effectiveness of pre-scan preparation [12]. These methods simulate the conditions of the procedure, increasing patient comfort and reducing anxiety.

This study evaluated the effectiveness of an early version of specialized training software that includes an integrated camera and



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motion recognition algorithms. The software is designed to reduce movements during the scan and increase the comfort level and familiarity of adult participants with real MRI conditions. Unlike previous studies, our research focuses on integrating advanced motion recognition algorithms and real-time feedback, allowing for a more accurate assessment of patient readiness for the procedure.

Effective feedback mechanisms are essential for the successful conduct of MRI scans, helping patients understand and correct their movements during the procedure. MRI software and simulators help patients adapt to noise and confined spaces, improving their psychological preparation and providing medical staff with the ability to assess patient readiness. Observing the patient's reaction to the simulated MRI conditions minimizes the risk of movement, improving image quality and enhancing the diagnostic value of MRI scans, ensuring timely diagnosis.

2 Methodology

We conducted an empirical study to evaluate the effectiveness of a motion detection application within an MRI simulator, focusing on an individual's readiness and performance during MRI procedures. For testing this software, a manually crafted simulator, similar in size and shape to a closed-type MRI machine, was created. The study included the following steps.

- (1) Preparation of the MRI machine simulator, including camera setup and connection to a computer with the necessary software installed.
- (2) Recruitment of volunteers to participate in the experiment.
- (3) Conducting MRI simulation sessions for participants to adapt to the sounds and sensations of the confined space.
- (4) Analysis of data obtained from the camera and software to study the behavior and reactions of participants during the simulation.

2.1 Materials and equipment

The study was conducted using a prototype of the MRI simulator, which was equipped with dedicated software for movement detection.

- (1) **Software:** Specially developed software was utilized for processing images from the MRI simulator's camera. It comprises two primary components: one for capturing camera frames and the other for their processing. Upon frame capture, motion analysis is conducted, and in the case of motion detection, information is stored and statistics updated. Upon completion of frame processing, camera resources are released for subsequent use. The software also provides functionalities for converting color spaces to enhance visualization and utilizes complex analysis methods for efficient motion detection. Figure 1 shows the view of several software windows, intended for MRI operators to monitor and adjust the system.
- (2) **MRI Simulator:** A compact simulator was developed, representing a model that mimics conditions comparable to those of a real MRI machine. This design was created to form an enclosed space that would replicate the atmosphere of an actual MRI examination. Key features included a curved roof

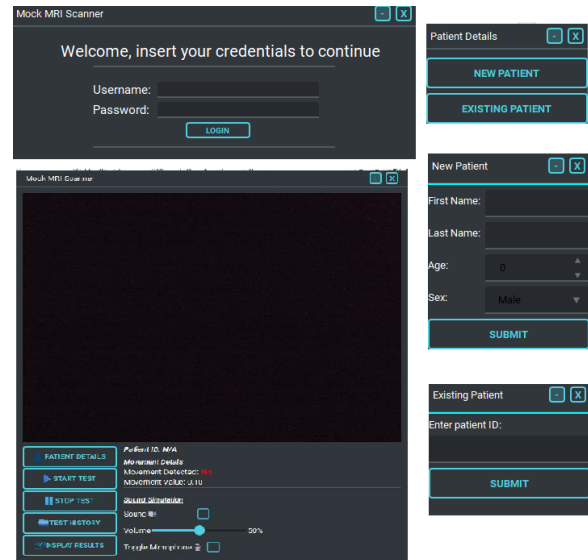


Figure 1: Operator Interface for MRI Simulator Monitoring and Adjustment

with a radius similar to that of a real MRI machine. Additionally, the construction integrated a camera and speakers to capture video images and reproduce corresponding sound effects encountered during MRI procedures. This solution ensured a realistic simulation of conditions akin to those experienced during a genuine MRI examination, making the sensations from the simulator as close as possible to a real MRI.

(3) Equipment Used for the Study

Main Equipment:

- **Personal Computer:** Used for computations, equipment control, and data processing.
- **Camera:** Required for real-time video capture during the experiment.
- **Microphone:** Utilized for recording sound signals and communication with experiment participants.

Development and Data Processing Tools:

- **Visual Studio and PyCharm:** Used for software development, system management, and data processing.
 - **NoSQL Database:** Employed for storing and managing experiment data, including session and participant information.
 - **GitHub:** Utilized for storing and managing project source code and documentation.
 - **OpenCV:** Computer vision library used for processing and analyzing images captured by the camera.
 - **PyAudio:** Library used for processing audio data and implementing communication functions with participants.
 - **Custom-developed Motion Detection Software:** Includes components for capturing and processing frames, enabling efficient motion detection and updating statistics.
- All interactions of the utilized equipment and tools are illustrated in Figure 2.

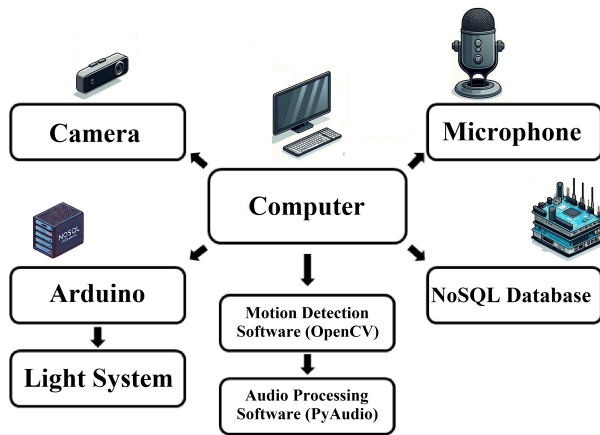


Figure 2: Equipment and Tools Used for Development and Data Processing

2.2 Motion Detection Algorithm

Farneback [10] optical flow algorithm was utilized for motion detection during the procedure. The Farneback algorithm identifies moving objects against static backgrounds, making it useful for analyzing complex dynamic scenes in real-time. The algorithm analyzes pixel intensity changes between consecutive video frames. First, frames are captured for analysis, and then pixel intensity changes are measured. Pixels with significant intensity changes and high vector speeds are identified as moving. The algorithm analyzes intensity changes between consecutive frames and calculates the speed and direction of each pixel, determining its new position in the next frame. Based on these changes, the Farneback optical flow equations estimate the speed and direction of each pixel, determining its new position in the next frame.

Post-processing methods are applied to eliminate noise and improve the accuracy of moving object detection, achieving precision levels of 95-98%.

These methods include converting the frame to grayscale to reduce processing complexity, applying blurring to reduce noise, and using a threshold to determine significant movement.

In the software interface, the result of the algorithm's operation is visually displayed, as shown in Figure 3

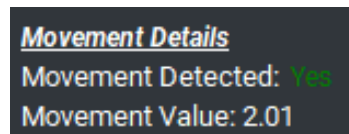


Figure 3: Movement Details

2.3 Participants

In the software interface, the result of the algorithm's operation is visually displayed, as shown in the image.

As part of our study, nine healthy volunteers were selected: two women and seven men aged 25 to 34 years (the mean age of participants was 29.1 years, the standard deviation was 2.73 years). All participants were students who participated voluntarily. The selection criteria included the absence of serious medical issues or

chronic conditions that could skew the results of our experiment. Before inclusion, each prospective participant was thoroughly briefed on the study's objectives, involvement procedures, potential risks (such as the possibility of claustrophobia attacks), and benefits (familiarization with the procedure and undergoing MRI). The experiment was approved by the college's ethics committee, which ensured compliance with all ethical norms and standards. Before the experiment, all participants provided written consent, confirming their willingness to participate and familiarize themselves with the research procedures.

2.4 Experimental Procedure

The experiment consisted of three stages presented in Figure 4 and detailed below.

1. Preparation

1.1 Participant guidance and explanation Briefing the participants on the study, explaining the procedures, and addressing any questions they might have.

1.2 Consent form Participants were then asked to sign consent forms.

1.3 Questionnaire After obtaining the consent form, participants filled out a questionnaire to gather demographic information.

2. Baseline test

2.1 Mock test In the second stage, participants entered a simulated MRI environment and underwent a mock test without any prior training. The test duration was 10 minutes. This served as the baseline test.

2.2 Data Collection The participants' movements were recorded, and the results were saved for subsequent analysis.

3. Post-training test

3.1 MRI preparation Preparing the participants for another round of testing in the MRI simulator.

3.2 Mock MRI test with guidance Participants then underwent a second mock MRI test, this time with specific training and guidance provided to minimize their movements during scanning. Similar to the baseline test, the post-training test lasted 10 minutes.

3.3 Data Collection The participants' movements after the training were recorded, and the results were saved for comparison with the baseline test.

3.4 Data analysis Data analysis was conducted to assess the effectiveness of the training. Participants' movements were recorded before and after the training, and the results were compared to identify improvements.

During both the baseline and post-training tests, the camera was directed at the upper part of the participants' bodies to observe their movements. Participants spent 10 minutes in the MRI simulator for each test. The software captured data on their movements, which were then analyzed to compare the number of movements between the baseline and post-training tests.

3 Results

The study provided data on participants' movements in a simplified simulated MRI environment. The results were evaluated using specialized software described in previous sections that analyzed

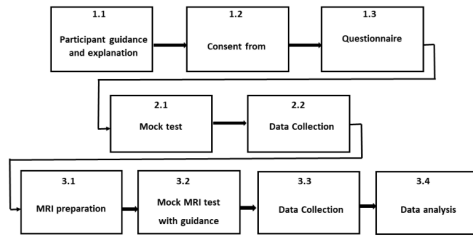


Figure 4: Stages of the Experimental Procedure for Assessing MRI Simulator Effectiveness.

and recorded the number of movements detected by the camera during each test. Only significant movements exceeding the threshold value were included in the statistics. The threshold value was determined from the mean magnitude of the optical flow vectors in the initial frames. Specifically, the code calculated the mean magnitude of the first 800 vectors and added a small value (0.01) to ensure the threshold was slightly above the average. The collected data included unique participant IDs, gender, and the number of movements detected during each test. Information about the recorded movements of participants is presented in Table 1.

Participant	Detected Movement Before Training	Detected Movement Post-Training	Improvement
1	3 times	0 times	Yes
2	14 times	5 times	Yes
3	0 times	0 times	No
4	6 times	4 times	Yes
5	0 times	0 times	No
6	1 times	0 times	Yes
7	0 times	0 times	No
8	4 times	3 times	Yes
9	7 times	9 times	No

Table 1: Comparison of Participants' Movements Before and After Training

The analysis of results focuses on the interpretation and evaluation of the obtained data using descriptive analysis. Results showed that three participants remained motionless during both tests, three reduced their movements in the second test compared to the first, two participants did not move at all during the second test, and one participant increased the number of detected movements. No difference was observed between male and female participants. A graphical comparison of the number of movements in the basic test and after training for all participants is presented in Figure 5.

To evaluate the effectiveness of the training, we conducted a statistical analysis comparing the number of detected movements before and after the training. We used the Wilcoxon signed-rank test [13], a non-parametric test for comparing two related samples. The results of the analysis showed a significant reduction in the number of movements after the training: the mean number of movements before the training was 10, while after the training, it was 4. The statistical significance was established at the $p < 0.05$ level.

4 Limitations

A small sample size and lack of diversity among participants can significantly reduce the reliability and generalizability of the study's

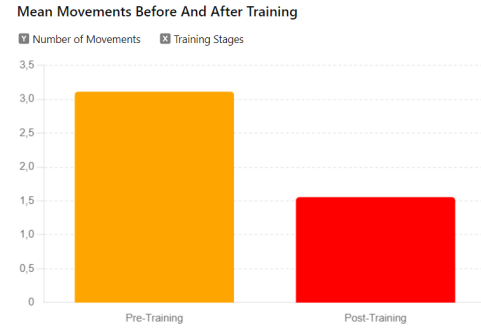


Figure 5: Average Number of Movements Before and After Training

results. Another important limitation is the potential behavioral differences between child and adult participants, which may affect the conclusions. It is also essential to consider the use of a simulator, which may not fully replicate real magnetic resonance imaging (MRI) conditions, potentially leading to discrepancies in the data.

Additionally, if the effect of training and familiarization with the procedure was not minimized or was interrupted before achieving a full positive outcome, subsequent MRI scans might show improved quality as participants become familiar with the procedure. This familiarity could reduce stress levels and behavioral artifacts that could influence the scanning results.

5 Conclusions and Future Research

Descriptive analysis of the data showed that the training positively influenced the participants' ability to minimize movements in the MRI simulator. Data analysis revealed that participants' movements decreased after training, confirming the effectiveness of the provided instructions.

In our future research, we plan to increase the number of participants and include various groups, including children and adolescents, to increase sample diversity. We also plan to conduct additional tests to evaluate the impact of participants' emotional states on their movements during scanning. Additionally, we are updating the software to implement a feedback mechanism that will allow participants to receive information about their movements and improve their training. Upon completion of testing with the next group of participants, we plan to conduct real MRI examinations to fully assess the effectiveness of the training before the MRI scan. We also aim to enhance the MRI simulator to more accurately resemble an actual MRI machine and better replicate real procedure conditions.

Furthermore, we plan to conduct data analysis to identify the optimal training methods that most effectively reduce movements, as well as evaluate the long-term effects of the training on movement reduction during subsequent MRI procedures.

References

- [1] Committee on Drugs American Academy of Pediatrics. 2006. Guidelines for Monitoring and Management of Pediatric Patients During and After Sedation for Diagnostic and Therapeutic Procedures. *Pediatrics* 118, 6 (2006), 2587–2602. <https://doi.org/10.1542/peds.2006-2780>
- [2] Jonathan Ashmore, Jerome Di Pietro, Kelly Williams, Euan Stokes, Anna Symons, Martina Smith, Louise Clegg, and Cormac McGrath. 2019. A Free Virtual Reality

- Experience to Prepare Pediatric Patients for Magnetic Resonance Imaging: Cross-Sectional Questionnaire Study. *JMIR Pediatrics and Parenting* 2, 1 (2019), e11684. <https://doi.org/10.2196/11684>
- [3] Kristin Barañano and Irina Burd. 2022. CNS Malformations in the Newborn. *Maternal Health, Neonatology and Perinatology* 8, 1 (Jan. 2022). <https://doi.org/10.1186/s40748-021-00136-4>
- [4] Charles J. Coté, Stephen Wilson, and The Work Group on Sedation. 2019. Guidelines for Monitoring and Management of Pediatric Patients Before, During, and After Sedation for Diagnostic and Therapeutic Procedures: An Update. *Pediatrics* 143, 6 (2019), e20191000. <https://doi.org/10.1542/peds.2019-1000>
- [5] William A. Edelstein, Mahadevappa Mahesh, and John A. Carrino. 2010. MRI: Time Is Dose—and Money and Versatility. *Journal of the American College of Radiology* 7, 8 (2010), 650–652. <https://doi.org/10.1016/j.jacr.2010.05.002>
- [6] NHS England. 2024. *Diagnostic imaging reporting turnaround times*. <https://www.england.nhs.uk/long-read/diagnostic-imaging-reporting-turnaround-times/>
- [7] IMAIOS. 2024. MR image quality and artifacts: Motion and Ghosting remedies. <https://www.imaio.com/en/e-mri/image-quality-and-artifacts/motion-and-ghosting-artifacts-remedies> Accessed: 2024-07-16.
- [8] Zachary Munn and Zoe Jordan. 2013. Interventions to Reduce Anxiety, Distress, and the Need for Sedation in Pediatric Patients Undergoing Magnetic Resonance Imaging: A Systematic Review. *Journal of Radiology Nursing* 32, 2 (2013), 87–96. <https://doi.org/10.1016/j.jradnu.2012.08.003>
- [9] Alalfy M. Nagy Raafat, M. 2021. Cite this article. *Egyptian Journal of Radiology and Nuclear Medicine* 52, 85 (Jan. 2021). <https://doi.org/10.1186/s43055-021-00465-6>
- [10] Sourav Dey Roy and Mrinal Kanti Bhowmik. 2020. A Comprehensive Survey on Computer Vision Based Approaches for Moving Object Detection. In *2020 IEEE Region 10 Symposium (TENSYP)*. 1530–1534. <https://doi.org/10.1109/TENSYP50017.2020.9230869>
- [11] Maic Masuch Stefan Liszio. 2017. Virtual Reality MRI: Playful Reduction of Children's Anxiety in MRI Exams. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 127–136. <https://doi.org/10.1145/3078072.3079713>
- [12] Kim L. et al. Suzuki A., Yamaguchi R. 2023. Effectiveness of mock scanners and preparation programs for successful magnetic resonance imaging: a systematic review and meta-analysis. *Pediatric Radiology* 53 (2023), 142–158. <https://doi.org/10.1007/s00247-022-05394-8>
- [13] Frank Wilcoxon. 1945. Individual comparisons by ranking methods. *Biometrics bulletin* 1, 6 (1945), 80–83. <https://doi.org/10.2307/3001968>